"I've Seen How This Goes": Characterizing the Diversity of LLM Generations and Human Writing via Progressive Conditional Surprise

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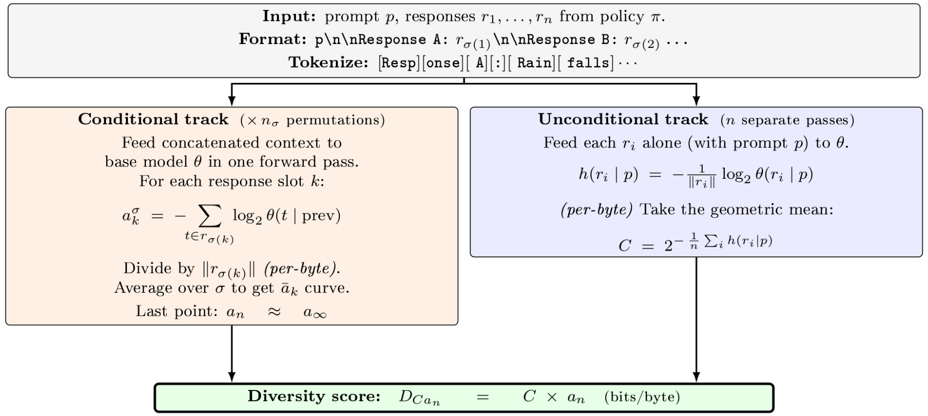
Paper

In-context learning of pretrained models can be used to directly measure the diversity of LLM generations and human writing.

$$D_{Ca_n} = C \times a_n$$

plausibility-weighted residual diversity (bits/byte)

A per-byte score read off a base model's per-token log-probabilities, with **no embedding model, no reference corpus, and no human labels.**



The same pipeline scores AI samples and human-written response sets.

Near SentBERT On Tevet & Berant's human-grounded McDiv, $C \times a_n$ tracks human diversity judgements, 3.7% to 21% behind SentBERT in OCA.

0.481 → 0.281 Decan drops monotonically across OLMo-2-7B post-training, detecting diversity loss.